SMART INDIA HACKATHON 2026

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- **Problem Statement ID –** SIH26262
- **Problem Statement Title –** Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** [ ]
- **Team Name (Registered on portal) –** Infera



ApprovalTrack · *Rules that run themselves. Decisions that stay human.*

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ApprovalTrack

Approval Workflow, Compliance & Alert Automation for AICTE

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"Rules that run themselves. Decisions that stay human."

THE PROBLEM TODAY

- Pendency, EVC visits, undertakings and lease renewals are chased by phone and email
- Deadlines and expiring documents slip because nobody is alerted in time
- No single view of who owns the next action, by when, and why

1 ROUTE → **2 CLOCK** → **3 ESCALATE** → **4 WATCH** → **5 PROVE**

- **1 ROUTE** — Versioned policy classifies each application to one of 5 routes; no match → an officer classifies
- **2 CLOCK** — Every stage and every institution response gets a due date, with the rule behind it
- **3 ESCALATE** — Reminders 30 / 7 / 1 days before, overdue → case officer → bureau. Only notifies
- **4 WATCH** — Approval conditions become tracked obligations: evidence, verification, renewal
- **5 PROVE** — Hash-chained audit: every action, actor, reason and document version

How it addresses the problem

1. **One action panel per case:** owner, stage, due date and next permitted step
2. **Deadlines are chased by the system,** not by staff; escalations go up the ladder
3. **Returns and re-scrutiny keep full history** (attempt 1, attempt 2 …)
4. **Rules change by publishing a new version;** running cases stay on theirs

Innovation & uniqueness

Obligation & evidence register

Decision → condition → obligation → deadline → evidence → verification → closure, all linked

Policy impact preview

See which routes and deadlines change before a new rule version is published

The system never decides

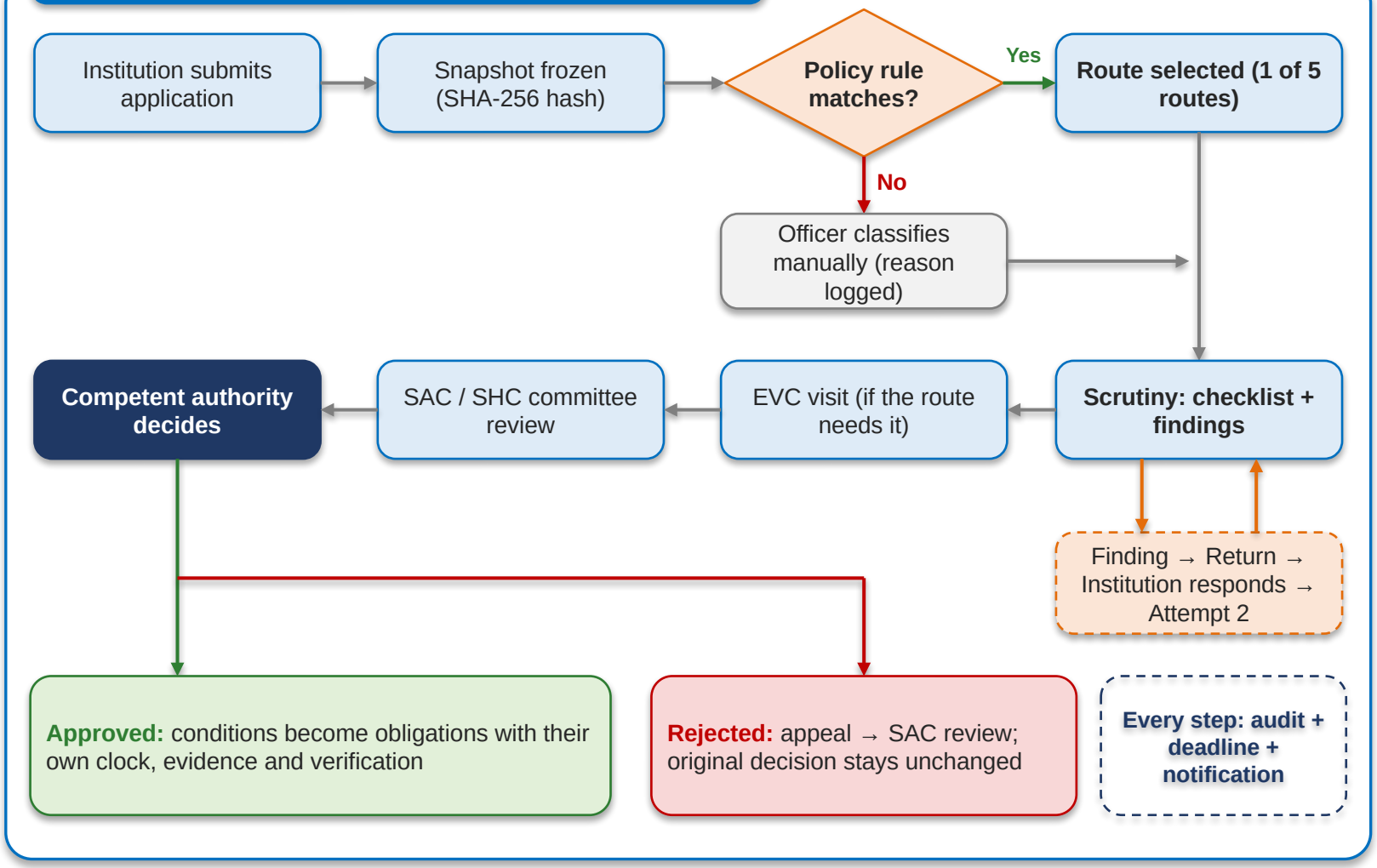
Only an authorised officer records approve / reject. Escalation only notifies

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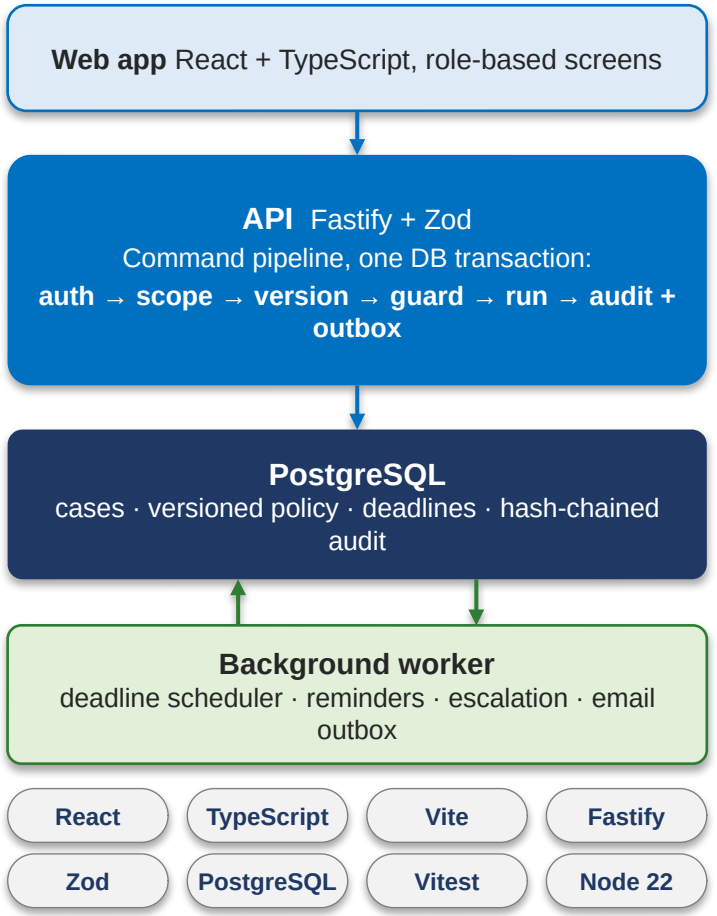
TECHNICAL APPROACH

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Methodology: life of an application (flowchart)



Architecture & stack



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FEASIBILITY AND VIABILITY

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99	automated tests passing against a real PostgreSQL database
8	integrated modules in one platform, one login, one record
5	approval routes running from a versioned, validated policy
₹0	licence cost: open-source stack, runs on one laptop, offline

The final platform: 8 modules working as one system

- **Secure login & role-based access**
- **Smart routing & full workflow**
- **Returns, re-scrutiny & appeals**
- **Deadlines, reminders & escalation**
- **EVC scheduling & conflict check**
- **Documents & obligation tracking**
- **Live dashboards & reports**
- **Tamper-proof audit & policy preview**

Challenge / risk	Strategy to overcome it
Rule accuracy vs APH 2024-27	Rules are versioned data with a source label (APH_CLAUSE / ADMIN_INSTRUCTION / DEMO_ASSUMPTION); domain review before publishing
Integration with the AICTE portal	Integration adapter with external-ID mapping and reconciliation, ready to plug into AICTE's authorised API
Adoption by officers & institutions	Role-based screens; every case opens on its action panel: what is pending, who owns it, by when
Scheduler downtime, duplicate alerts	Deadlines live in the database; worker recovers on restart, sends only the current step, suppresses duplicates (tested)

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IMPACT AND BENEFITS

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Target audience: who benefits

Institutions

See what is pending, why and by when; answer findings online; reminders before every deadline

AICTE officers & regional bureaus

One queue with owner, stage and due date; escalations reach them only when needed

Expert committees (EVC / SAC / SHC)

Assigned cases arrive with checklist, findings and the full history of attempts

Competent authority

A complete, verifiable trail behind every recommendation; the decision stays theirs

PS expected outcome	Today (manual)	With ApprovalTrack
Less manual follow-up	Staff chase files by phone and email	Reminders 30 / 7 / 1 days before and an escalation ladder run on their own
Fewer missed deadlines & expired documents	Dates kept in spreadsheets and inboxes	Durable clock per stage; pauses on return; document validity tracked apart from upload
Visible pendency by region & stage	Status scattered across offices	Scoped queues show every pending case with owner, stage and due date
Accountability	Hard to reconstruct who did what	Tamper-evident, hash-chained audit with a one-click integrity check

Social: transparent, predictable process for institutions and students

Economic: staff time moves from chasing files to deciding; zero licence cost

Environmental: paperless files and fewer physical follow-up visits

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RESEARCH AND REFERENCES

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References

1. **AICTE Approval Process Handbook 2024-27**
   aicte.gov.in/sites/default/files/APH Final.pdf
2. **AICTE APH addenda / corrigenda 2025-26 and 2026-27**
   EVC and committee definitions; official copy to be verified
3. **PIB: launch of the Approval Process Handbook 2024-27**
   pib.gov.in/Pressreleaseshare.aspx?PRID=1983013
4. **National Single Window System (NSWS) FAQs**
   nsws.gov.in/faqs: final decisions stay with the authority
5. **SIH 2026 Problem Statement SIH26262 (AICTE)**
   sih.gov.in: expected outcomes and evaluation criteria
6. **Published EoA, SAC and SHC formats**
   Studied for stage names, fields and decision records
7. **PostgreSQL documentation: row locking**
   SELECT … FOR UPDATE SKIP LOCKED for a durable deadline queue

What we studied → what we built

Route table (APH Table 2.3)	5 configurable routes
Hearing & appeal process	Appeal as a linked workflow
No edits after submission	Frozen snapshot + amendments
EVC / SAC / SHC definitions	Committee stages and queues

Honest by design

APH_CLAUSE | ADMIN_INSTRUCTION | DEMO_ASSUMPTION

Every rule carries its source label; no demo rule is shown as an AICTE rule.

All demo data is synthetic. No live AICTE connection is claimed.